

Project Report

Project Title: PWM LED Dimmer using Arduino

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1. Objective:

The objective of this project is to control the brightness of an LED using Pulse Width Modulation (PWM). A potentiometer is used to vary the brightness by changing the PWM duty cycle generated by the Arduino Uno.

2. Components Used:

- Arduino Uno
 - LED
 - 220 Ω Resistor
 - Potentiometer (10 k Ω)
 - Jumper Wires
 - Wokwi Arduino Simulator
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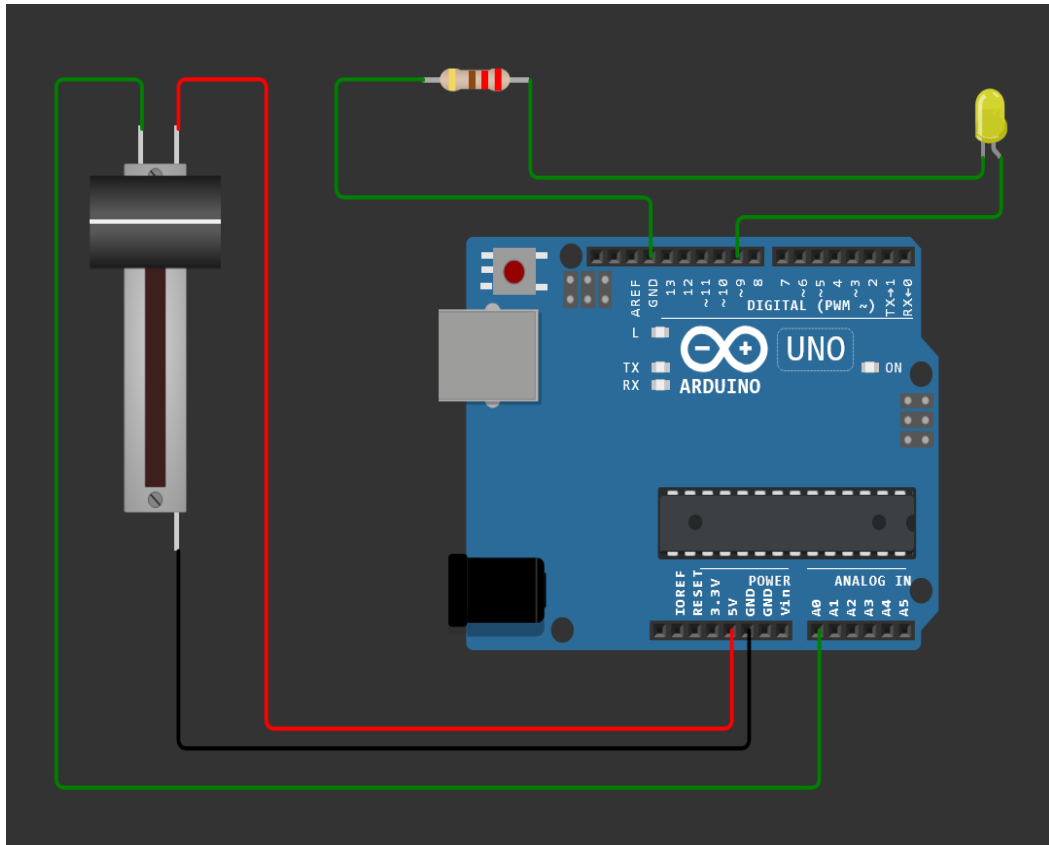
3. Circuit Connections:

Potentiometer:

- VCC $\rightarrow$ 5V
- GND $\rightarrow$ GND
- SIG $\rightarrow$ A0

LED:

- Anode (+) $\rightarrow$ Pin 9 (PWM Pin)
- Cathode (–) $\rightarrow$ 220 Ω Resistor
- Resistor $\rightarrow$ GND



4. Working Principle:

The potentiometer provides an analog voltage to the Arduino through pin A0. The Arduino reads this value using the `analogRead()` function, which produces a value between 0 and 1023.

Since PWM output requires a value between 0 and 255, the `map()` function converts the input range (0–1023) into the output range (0–255).

The `analogWrite()` function then generates a PWM signal on pin 9. By rotating the potentiometer, the PWM duty cycle changes, which increases or decreases the brightness of the LED.

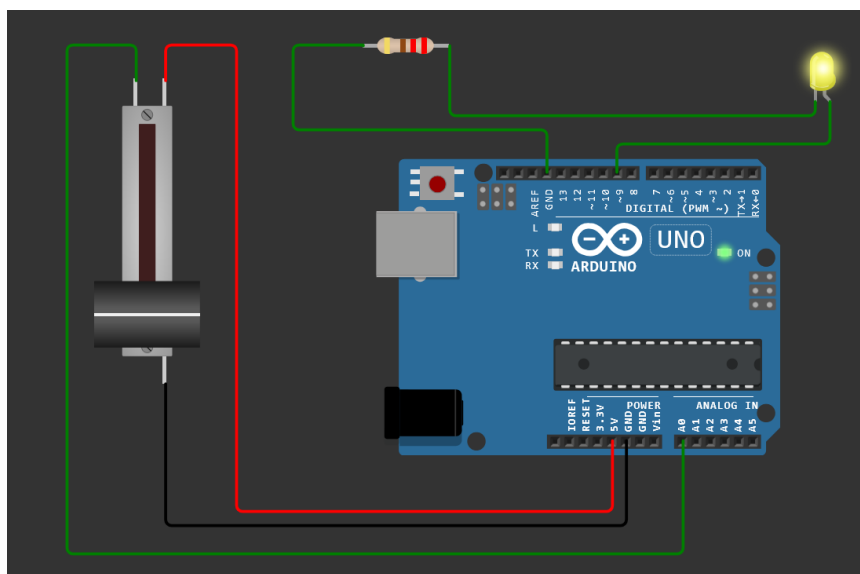
5. Arduino Program:

```
const int potPin = A0;  
const int ledPin = 9;  
  
void setup() {  
  pinMode(ledPin, OUTPUT);  
}  
  
void loop() {  
  int sensorValue = analogRead(potPin);  
  int brightness = map(sensorValue, 0, 1023, 0, 255);  
  analogWrite(ledPin, brightness);  
}
```

6. Output:

The LED brightness changes according to the position of the potentiometer.

- Potentiometer at minimum → LED dim.
- Potentiometer at maximum → LED bright.



7. Applications:

- LED brightness control
 - Home lighting systems
 - Energy-efficient lighting
 - Industrial lighting control
 - Educational demonstration of PWM
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8. Advantages:

- Simple and low-cost circuit
 - Easy to understand and implement
 - Low power consumption
 - Smooth brightness control
 - Suitable for beginners learning Arduino and PWM
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9. Future Scope:

- Control multiple LEDs independently.
 - Add an OLED/LCD display to show brightness percentage.
 - Control LED brightness using a mobile application via Bluetooth.
 - Use IoT for remote brightness control.
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10. Conclusion:

The PWM LED Dimmer project was successfully implemented using Arduino Uno and simulated using the Wokwi platform. The project demonstrated the working of Pulse Width Modulation and analog input processing. It is a simple and effective project for understanding PWM-based LED brightness control.
